ERIK HENRIQUE ALVES CUNHA

SENIOR FRONTEND ENGINEER

React · Next.js · Angular · TypeScript · Node.js · Python · AWS · AI Agents · RAG · LLMs · Evals · Design Systems · Micro-Frontends

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PROFESSIONAL SUMMARY

Senior Software Engineer with 8+ years building production web software across AI-native SaaS, fintech, payments, healthcare and e-commerce, including 8M+ monthly users at Nike Brazil and Centauro. I specialize in React, Next.js, TypeScript and frontend architecture: design systems, micro-frontends, server-side rendering (SSR), static site generation (SSG), incremental static regeneration (ISR) and backend-for-frontend (BFF) layers, with full-stack work in Node.js, Fastify, PostgreSQL, Python and AWS.

At Raylu, an AI deal origination platform used by 50+ venture, growth and private equity funds, I build a reusable data grid on TanStack, versioned URL-backed view state and accessible components, shipping behind a production A/B that cut Cumulative Layout Shift (CLS) from 0.166 to zero and one route's Largest Contentful Paint (LCP) from 2,544 to 1,308 ms. At Betsson I built payment experiences on a PCI-DSS platform processing 40M+ transactions a year across 15+ regulated markets.

I also build applied AI: a streaming Vercel AI SDK endpoint behind a CI eval harness, with an LLM judge calibrated against my own labels at a 0.85 agreement floor, jailbreak resistance gated at 1.0 and a per-job cost cap.

TECHNICAL SKILLS

Frontend & Product Architecture: React, Next.js, TypeScript, JavaScript (ES6+), HTML, CSS, SCSS, TanStack Table and Query, Plate, Tailwind CSS, Radix UI, styled-components, design systems and Storybook, micro-frontends, Web Components (StencilJS), async state machines with RxJS, NgRx and Redux Toolkit, URL-synchronized view state, grid virtualization, WAI-ARIA, i18n, technical SEO

Testing, Quality & Release Engineering: Core Web Vitals: LCP, INP, CLS; Web Content Accessibility Guidelines (WCAG) 2.2 AA; Playwright, Jest, Vitest, React Testing Library, Chromatic, MSW, WireMock, Stryker mutation testing, Nx monorepos, Husky, continuous integration and delivery (CI/CD) on GitHub Actions and AWS CodePipeline, Docker, NGINX, AWS (S3, CloudFront), Sentry

Backend-for-Frontend (BFF), APIs & Data: Node.js, Fastify, Express, Python, REST APIs, JSON:API, BFF layers, webhooks, WebSocket streaming (STOMP/SockJS), PostgreSQL, TypeORM, Drizzle ORM, MongoDB, MS SQL Server, SQLite, Redis, ETL pipelines

AI-Native Product & Agent Engineering: Vercel AI SDK, Anthropic API, OpenAI, Model Context Protocol (MCP) servers, Zod structured outputs, LLM-judge eval harnesses calibrated against human labels, per-job cost caps, prompt-injection defense, Claude Code and Codex skills and lifecycle hooks, spec-driven development, quality gates that fail closed

PROFESSIONAL EXPERIENCE

Raylu | Senior Software Engineer (Contract via Torc) — Jun 2026 – Present

New York, US | Remote | Private-markets fintech | B2B SaaS, Series A | 50+ funds holding $500B+ in assets

- Rebuild the investor workflow screens of a private-markets platform (company, pipeline, comparison, outreach), giving funds holding $500B+ in assets one surface to shortlist, compare and contact founders
- Ship the data-platform migration behind a production A/B across four Next.js routes, cutting controls without an accessible name from 55 to 1, CLS from 0.166 to zero and one route's LCP from 2,544 to 1,308 ms, and Lighthouse performance from 79 to 96
- Engineer the table engine, a client mode on TanStack Table paired with a server mode on React Query (cancellable requests, deterministic caching, faceted filter counts, shape validation between them), at 94.7% line and 85.4% branch coverage against library baselines of 85.4% and 65.8%
- Design the versioned view-state contract the React pages share, encoding sort, filters, search, columns and grouping in URL-safe TypeScript codecs, so every shared or stale link resolves to a working view
- Build the shared outreach editor on Plate and React, holding personalization as document structure so authors see which fields resolve before sending

- Virtualize flat and grouped rows over a real table element, keeping native accessibility, with the grid's WAI-ARIA roles, roving-tabindex toolbars and live-region announcements specified in the design docs before code
- Write design documents for the data platform and editor before building (engine choice, migration order, rollback windows), then move the list pages onto the engine in production, fixing shared filtered links that returned unfiltered data
- Deliver Fastify, TypeORM and PostgreSQL changes: a hand-written migration with a rehearsed backfill and tested rollback, and an EXISTS-subquery fix that raised one owner's team-filter matches from 14 to the correct 75

Stack: Fastify, TypeORM, PostgreSQL, TypeScript, React, Next.js 14, TanStack, Plate, Tailwind CSS, Radix, Nx, Storybook, Chromatic, Jest, React Testing Library, Playwright

Betsson Group | Senior Software Engineer — Mar 2025 – May 2026

Malta | Hybrid | Online gaming and payments | B2C, Nasdaq Stockholm-listed | €1.197B revenue, 31.1M customers

- Built the cashier micro-frontend platform every deposit and payout across the group's brands passes through, isolating brand orchestration from payment primitives on 40M+ transactions a year
- Implemented the shared payment primitives as framework-agnostic StencilJS Web Components driven by white-label configuration across 5+ brands, 8+ languages and 10+ currencies, zero-decimal ones included
- Designed the multi-step deposit and withdrawal flows as async state machines in TypeScript with RxJS and NgRx, with recovery paths for failed steps, across 15+ regulated jurisdictions
- Integrated Trustly open-banking deposits for the Nordic markets and Yape payments for Peru (about 16M monthly users) as StencilJS components configured per country and brand, tested against WireMock and Playwright mocks
- Tested the KYC and payment flows with Jest, Playwright and WireMock against mocked providers, so provider outages and API changes did not break the suite, and enforced standards with Husky hooks and secret scanning
- Built a 12-agent AI-assisted engineering system on GitHub Copilot spanning generation, review, debugging, testing and architecture validation, adopted across the team with every automated action reviewable

Stack: TypeScript, Nx Monorepo, StencilJS, Web Components, Micro-frontends, RxJS, NgRx, Angular, Storybook, Jest, Playwright, WireMock, GitHub Actions, GitHub Copilot

Independent Consultant (Canon Medical Systems Brazil) | Senior Software Engineer — Mar 2023 – Mar 2025

Canada | Remote | Healthcare technology | B2B consulting | ¥580.6B net sales

- Rebuilt Canon Medical's hospital equipment service platform, replacing a legacy tool with a layered domain, data and presentation architecture for service orders, specialist schedules and field visits
- Cut time to interactive (TTI) about 52% via concurrent RxJS loading and change-detection tuning, the JavaScript bundle about 33% and the layout shell's CSS about 98% via lazy loading and PrimeFlex, lowering initial load about 25%
- Automated releases (Bash, SSH, NGINX): updates from about 5 minutes to about 45 seconds, rollbacks about 1 minute
- Upgraded the framework from version 17 to 19 and Nx from 17 to 20 in staged steps, moving 52 of 54 components to OnPush change detection with standalone components, signals and the new control flow
- Built specialist visit scheduling on Bryntum Scheduler, FullCalendar and the Google Maps API with travel-time checks, plus a 29-permission role-based access control (RBAC) model and a retrying, de-duplicating HTTP interceptor
- Built the operations dashboard in TypeScript and Chart.js, the customer-satisfaction module (survey form builder, WhatsApp message templates, satisfaction goals) and real-time WebSocket (STOMP/SockJS) notifications
- Held Lighthouse accessibility scores near 100 across production views, built to WCAG AA, alongside JWT authentication, RBAC and Husky and ESLint quality gates

Stack: TypeScript, Nx Monorepo, RxJS, Clean Architecture, WebSocket (STOMP/SockJS), Angular, PrimeNG, Chart.js, FullCalendar, Bryntum Scheduler, Google Maps API, PWA, Husky, Bash, SSH, NGINX, GitHub Actions

Grupo SBF (Nike Brazil, Centauro) | Software Engineer — Dec 2021 – Mar 2023

São Paulo, Brazil | Remote | Sports retail e-commerce | B2C, B3-listed | R$7.9B gross revenue (2022), 31.9% digital

- Rebuilt Nike Brazil's storefront, including the listing, product detail and search pages, behind a backend-for-frontend (BFF) gateway and cut over in a phased rollout
- Kept the Nike and Centauro storefronts, serving 8M+ monthly users, isolated inside the shared Nx monorepo with enforced module boundaries, without forking the codebase
- Cut page load about 32% on storefronts carrying R$2.5B of 2022 digital revenue through Next.js code splitting, dynamic imports, image optimization and caching, with Core Web Vitals about 12% better, led by LCP

- Lifted conversion about 10% through 20+ Google Optimize 360 A/B experiments with pre-agreed stop-loss criteria, and implemented technical SEO (schema.org data, sitemaps, canonical URLs, ISR for product pages)
- Built the Nike and Centauro design system with brand theming and Storybook, covered four revenue-critical checkout journeys with Jest, React Testing Library and MSW at about 70% coverage, and mentored four junior engineers

  Stack: React, Next.js, TypeScript, styled-components (SSR extraction), NextAuth, Sentry, Nx Monorepo, Webpack, SSR/SSG, ISR, BFF gateway, Storybook, Jest, React Testing Library, MSW, Google Optimize, Google Analytics

Encora (formerly Daitan), assigned to VMware | Frontend Developer — Jan 2021 – Dec 2021

Campinas, Brazil | Enterprise cloud software | B2B, IT services delivery for VMware | $12.85B revenue (FY2022)

- Engineered VMware Pathfinder, the evaluation portal behind 2.1M+ labs, where customers run live interactive labs in the browser and follow guided paths before buying
- Fed catalogue pages from CMS content, so copy changed without a deploy, localized the portal through i18n, and instrumented funnel analytics for lab starts, completions and assessments
- Held unit-test coverage at about 70% with Karma and Jasmine on the workflows that gate a lab launch, and shipped through AWS CodeBuild to S3/CloudFront with cache invalidation, per-environment config and error monitoring

  Stack: TypeScript, NgRx, RxJS, Reactive Forms, i18n, Angular, Karma, Jasmine, AWS CodeBuild, S3, CloudFront

Zup Innovation (Itaú Unibanco) | Frontend Developer — Apr 2020 – Jan 2021

Campinas, Brazil | Bancassurance | Business insurance for Itaú Unibanco | 24.2M digital clients, 1.2M businesses (Dec 2020)

- Built micro-frontends to replace Itaú's legacy business insurance application journey by journey, packaged as custom elements and rendered at runtime by the bank's internal host platform
- Shipped each micro-frontend through its own AWS CodePipeline and CodeBuild pipeline to S3 and CloudFront, so it deployed on its own schedule instead of waiting for a full release of the web app
- Built the offer showcase for 4+ partner insurers (MetLife, Amil), driving each insurer's flow from configuration so a new insurer needed no frontend code
- Defined each micro-frontend's host contract (custom-element attributes in, DOM CustomEvents out) and built the quote, contracting and proposal tracking flows with Reactive Forms, validating CNPJ and covered lives

  Stack: TypeScript, Web Components (custom elements), RxJS, Reactive Forms, Angular, Angular Elements, Jasmine, AWS CodePipeline, CodeBuild, S3, CloudFront

Venturus — Feb 2019 – Apr 2020

Campinas, Brazil | Applied R&D | Non-profit institute, B2B client delivery | toll corridor at 850,000 vehicles a day

Frontend Developer — Nov 2019 – Apr 2020

- Developed the traffic-monitoring dashboards for CCR AutoBAn (now Motiva), whose 24/7 control centre watches 106 cameras and 36 traffic sensors along a 319.8 km corridor carrying 850,000 vehicles a day
- Built the live views for traffic volume, vehicle classification, average speed and lane occupancy, refreshing automatically from the 36 roadside sensors, with filtering in TypeScript and RxJS by segment, direction and time window
- Replaced the concession's spreadsheet billing with a browser-based workflow: CSV import and export in TypeScript, row-by-row validation (toll plaza, date, category and axles, amount) before saving, and a monthly close that locked edits and totalled gross revenue, equivalent vehicles and average tariff

  Stack: TypeScript, RxJS, NgRx, Angular, Karma, Jasmine, Jenkins

Full Stack Engineer — Feb 2019 – Nov 2019

- Built CGO Assessoria's customs-compliance platform end to end for an import-export consultancy active since 1989, cutting its reporting API from about 40s to under 1s and other endpoints from 2s to 200ms
- Built the Express REST API with input validation, central error middleware and paginated listings, writing multi-step changes in PostgreSQL transactions behind foreign keys and constraints, covered by API integration tests
- Modelled each foreign-trade process as a state machine through its regulatory stages (import licence, MAPA consent, release), with a document checklist per process type and deadline alerts
- Traced the slow reporting endpoint to how the ORM queried the data and fixed it by rewriting the queries and adding PostgreSQL indexes, instead of caching a slow read
- Migrated document-heavy workloads from MongoDB to PostgreSQL with a Node.js ETL pipeline, validating the cutover against both databases, with MongoDB read-only as the rollback path

- Generated the PDF and Excel reports server-side in Node.js, emailed clients when a process changed status, and secured the dashboard with JWT, an HTTP interceptor and route guards

  Stack: TypeScript, Node.js, Express, PostgreSQL, MongoDB, RxJS, Angular, Docker, Docker Compose, JWT, Sentry, Jenkins

MB Labs **Jul 2018 – Feb 2019**

Campinas, Brazil | Software house | B2B custom builds | education publisher across 10 countries, retail training

Software Engineer **Sep 2018 – Feb 2019**

- Architected Edições SM's offline-first learning app for Android, iOS, Windows, macOS and Linux, sharing about 80% of code, cutting desktop build time about 40%, and shipped to App Store, Google Play and desktop
- Built the in-app book reader and offline content pipeline with Cordova file I/O and SQLite (resumable downloads, local catalogue and reading position), and a sync that merges two devices' offline reading histories on reconnection

  Stack: TypeScript, Ionic, Electron, Cordova, SQLite, REST APIs, Angular

Full Stack Engineer **Jul 2018 – Sep 2018**

- Built the frontend and backend services for Hondana INDICA, the training chatbot used across Cia Hering's 800+ stores
- Built the Node.js dispatcher that fans one training push out to Firebase Cloud Messaging and Facebook Workplace, with retry and backpressure per channel, so one channel's failure never dropped a campaign
- Built the React chat interface and admin panel for training campaigns and per-store results, and made Node.js webhook ingestion idempotent on the message id after duplicate Workplace deliveries double-counted answers

  Stack: React, Node.js, Express, Webhooks, Firebase Cloud Messaging, MS SQL Server, AWS (EC2, RDS, S3, Route 53), NGINX

SELECTED PROJECTS AND OPEN SOURCE

- **react-google-maps-api (PR #3266)**: Fixed an infinite React re-render loop in the Directions API example of a library with about 1.8M weekly npm downloads: a callback prop with a new identity on every render cost the docs site 61K+ unwanted Google Maps API calls. Reproduced on a minimal StackBlitz; merged by the maintainer.
- **erikunha.dev**: Next.js 16 and React 19 portfolio on Vercel, RSC-first, with CI gates on LCP under 1.8s, INP under 200ms, 120KB gzipped JS per route and Lighthouse accessibility 100. Its streaming Anthropic endpoint (Vercel AI SDK) ships only if a 39-case eval corpus passes in CI against the live model, graded by an LLM judge calibrated on human labels at a 0.85 agreement floor, with correctness gated at 0.9, jailbreak resistance at 1.0 and a $2 per-job cost cap. Output guardrails reject injection at the request boundary and catch a leaked marker split across streamed chunks, with content-hashed prompt versioning recorded on every eval run and model-drift detection. An MCP server exposes the endpoint as a tool, and an agent-eval suite grades the coding agents that maintain it.
- **LAIbemage**: Open-source TypeScript CLI (MIT) classifying images across seven LLM vision providers, built as a study in auditable probabilistic pipelines: self-elicited confidence, escalation from a local model to cloud below a 0.70 threshold, two-provider consensus voting and a self-critique pass. Benchmarked on per-category precision and recall, p95 batch latency and normalized cost, gated at 5% regression across eight CI workflows.

EDUCATION

MBA (Lato Sensu), AI Engineering, Machine Learning and Deep Learning Expected Dec 2027

Instituto Infnet, Brazil (online) | Coursework: ML, deep learning, LLMs, retrieval-augmented generation (RAG), multi-agent systems, Python

Web Development Co-op Diploma 2023 – 2024

Cornerstone International Community College of Canada (CICCC), Vancouver, Canada

B.S. Information Systems 2015 – 2020

Centro Universitário Salesiano de São Paulo (UNISAL), Campinas, Brazil

ADDITIONAL INFORMATION

Certifications: Angular Developer Certification, certificates.dev (2024) · WES verified academic qualifications (2022)

Languages: Portuguese (native) · English (C1, IELTS 2023) · Spanish (B1) · French (A2)

Community: Organizer, DevOpsDays Campinas 2024: curated 10+ talks across DevOps, cloud infra, platform engineering and architecture. Ran the full speaker cycle: CFP launch, review, selection, program design, day-of and wrap. Coordinated with speakers, sponsors and co-organizers end to end.